

Topic 2.13 - Exponential and Logarithmic Equations and Inequalities

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Solve exponential equations and inequalities exactly or approximately.
- Solve logarithmic equations and inequalities while enforcing domain restrictions.
- Interpret solutions and thresholds in transformed functions and contextual models.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Solve $8^{x+1} = 4^{2x-1}$.

Question 2

Core skill**Calculator permitted**

Solve $7^{2x-3} = 20$ exactly and to three decimal places.

Question 3

Core skill

No calculator

Solve $4^x - 5(2^x) + 4 = 0$.

Question 4

Core skill

No calculator

Solve $3^{x+1} = 5^{x-2}$ exactly.

Question 5

Core skill

No calculator

Solve $\log_3(x^2 - 4) = 2$.

Question 6

Core skill

No calculator

Solve $\log x + \log(x - 9) = 1$.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

Solve $\ln(x + 4) = 2 \ln x$.

Question 8

Core skill

No calculator

Solve $2^{x-3} < 1/16$.

Question 9

Core skill

No calculator

Solve $\log_{1/3}(2x - 1) \geq -2$.

Question 10

Error analysis

No calculator

A student solves $\ln(x - 2) + \ln(x + 2) = \ln 5$ and keeps both roots of $x^2 - 4 = 5$. Correct the solution.

Question 11

Context threshold

Calculator permitted

A medication model is $A(t) = 180(0.78)^t$. Find when the amount first falls below 25 mg, with t measured in whole hours.

Question 12

Inequality

No calculator

Solve $5 - 2\log_3(x - 1) \leq 1$.

Section C: AP-style multiple choice**MCQ 1**

No calculator

The exact solution of $3^x = 11$ is:

- A. $x = 11/3$
- B. $x = \log_3 11$
- C. $x = 3^{11}$
- D. $x = \ln 3 / \ln 11$

MCQ 2

No calculator

Which candidate must be rejected when solving $\log(x - 2) + \log(x + 2) = 1$?

- A. $x = \sqrt{14}$
- B. $x = -\sqrt{14}$
- C. Both
- D. Neither

MCQ 3

No calculator

The solution of $(1/4)^x > 16$ is:

- A. $x > -2$
- B. $x < -2$
- C. $x > 2$
- D. $x < 2$

MCQ 4

No calculator

Which step is essential after solving a logarithmic equation?

- A. Differentiate the result
- B. Check candidates in the original domain
- C. Convert every value to a decimal
- D. Graph only the left side

Section D: Free-response reasoning**FRQ 1 - Mixed exact solutions**

No calculator

6 points

(a) Solve $9^x - 10(3^x) + 9 = 0$. (b) Solve $\log_2(x - 1) + \log_2(x - 7) = 4$. (c) State and apply all domain restrictions. (d) Verify the valid logarithmic solution.

FRQ 2 - Thresholds and inequalities

Calculator permitted

6 points

A model is $P(t) = 900(1.13)^t$, where t is measured in whole years. (a) Solve $P(t) \geq 2500$ over real t . (b) State the first whole year satisfying the inequality. (c) Solve $900(1.13)^t < 700$ and interpret the result. (d) Explain how the answers would change if the factor were 0.87 instead of 1.13.
